Inorganic Cumulative Exam
September 10, 2010

Describe a practical laboratory preparation for H_2 that does not involve a cylinder of hydrogen.

How can the above preparation be modified to yield D_2?

What is the volume of 5 g of H_2 at 22 °C and 762 torr?

What is the density of H_2 at 22 °C and 762 torr?

What are ortho and para hydrogen?

Reduction of water to give hydrogen is relatively fast at an electrode in an electrochemical cell but is very slow with a 1 electron reductant. Why?

What is a hydrogen economy and why is it not practical?

What is the bond order of H_2^+?

Complete the following:

$CaC_2 + H_2O =$

$CuSO_4 + H_2O =$

$CaO + H_2O =$

$SO_3 + H_2O =$

$P_4 + H_2O =$